Assistive Robotics

RESEARCH, DATA ANALYSIS & RECOMMENDATIONS
REGARDING ROBOTICS IN THE ELDERLY CARE CRISIS

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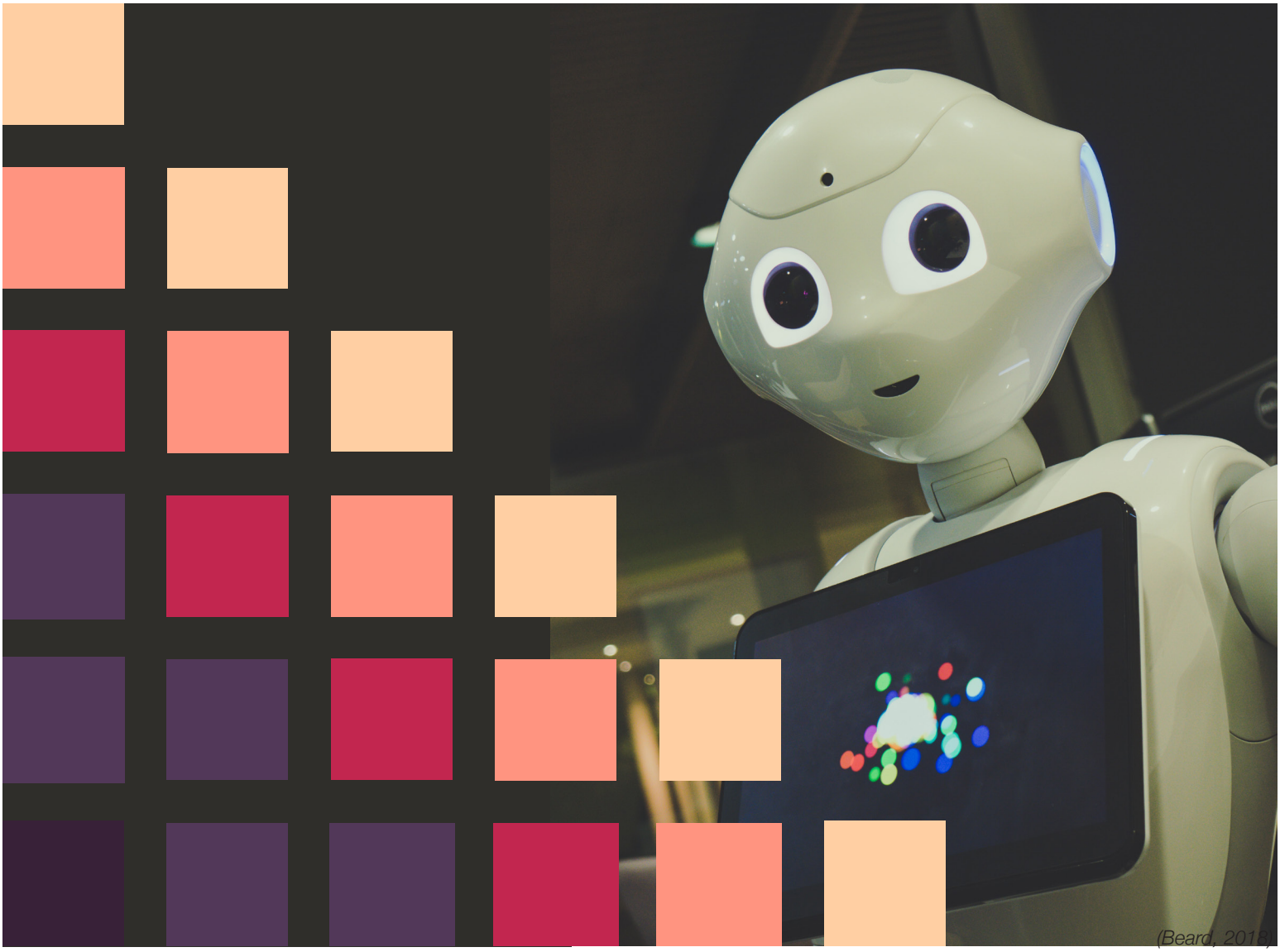
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Part One

EXPLORING ETHICAL & PRACTICAL ISSUES IN ROBOT COMPANIONS FOR THE ELDERLY



(Beard, 2018)

Robots in Today's World

A SNAPSHOT OF THE CURRENT MARKET

In 2008, when robotics across all industries began to become more relevant in everyday life, one of the ethical risks raised from this evolution was the care of children and the elderly (Sharkey, 2008). Since the 1980s, the robotics industry has made enormous strides to become both more affordable and available to the general public.

Countries such as Japan have introduced the Hybrid Assistive Limb Suit (HAL) to assist the elderly in walking. This suit multiplies the strength factor of its user by 2-10 times (Sharkey and Sharkey 2010). An assistive walking suit is just one example of the increase in availability of robotics worldwide. As the robotics industry grows, more and more companies are going to attempt to break into the industry. With a greater selection of robots, hospitals and elderly care centers are faced with the tough decision on where to allocate their money.

Gecko Systems International Corp has predicted that sales of their elderly care robots will reach $8.3 billion by 2014 (Sharkey and Sharkey 2010). Their CareBot robot is able to provide all the interactions that a human in-home nurse can at a one time cost. The main obstacle that the general public faces with the integration of these robots is the cost of each unit. The Remote Presence Robots (RP-6 and RP-7), made by Intouch Health started off at a listing price of about $80,000 (Sharkey and Sharkey 2010). While the initial price of many of these robots start off at a high price, as competition increases they will become more affordable and incorporated into more hospitals and outpatient homes.

As more and more robots make their way into healthcare, a big issue that manufacturers and healthcare professionals alike must face are tough questions relating to regulations and abiding by the medical codes that are enforced in each respective country. Potential issues that may need to be addressed in legislation include a feeling of isolation, and underdevelopment in young children who may not have the influence of an adult (Sharkey 2008). Even with these potential issues, no national or international legislation exists outside of negligence. As the future of both elderly care and healthcare as a whole trends towards robotic helpers, it will be imperative for governments and other policy-making institutions to create a code of ethics and rules for hospitals and other organizations to abide by.

$80,000

base list price of the Remote Presence Robots by Intouch Health



(V., 2018)

THE INDUSTRIES BEHIND THE ROBOT REVOLUTION

The most influential industries behind the motivation to use robots in post-operative care or as companions, are the robotic tech industry, the medical device industry, and the service robotics industry.

Improvements in the daily lives of the elderly can be seen with innovations from robotic tech companies. Although there is not the companion element that caregiving robots from the service robotics industry have, innovations from companies such as Cyberdene allow for an increase in strength by anticipating movements through nerve signals. Walking Assist machines from Honda also fall into this category of industries who are advocating for technological solutions to assist in elderly care. These inventions are more sophisticated than devices such as wheelchairs and can be used to persuade doctors to switch to devices like these in order to preserve patient dignity for those who may find being in a wheelchair embarrassing (Sharkey, 2012).

Robotic tech companies can help improve the lives of those who need daily care and those who give daily care by allowing for unpleasant tasks for both groups, such as bathing and toileting, to be assisted by robots. For example, a wheelchair that can be summoned by the patient could aid in a greater level of autonomy and could relieve both the caregiver and the patient of tasks a person would rather do alone. This type of assistance from the medical device industry could help convince doctors to switch to their devices because of this elimination of aiding with unpleasant tasks which is also a time saver (Sharkey, 2012).

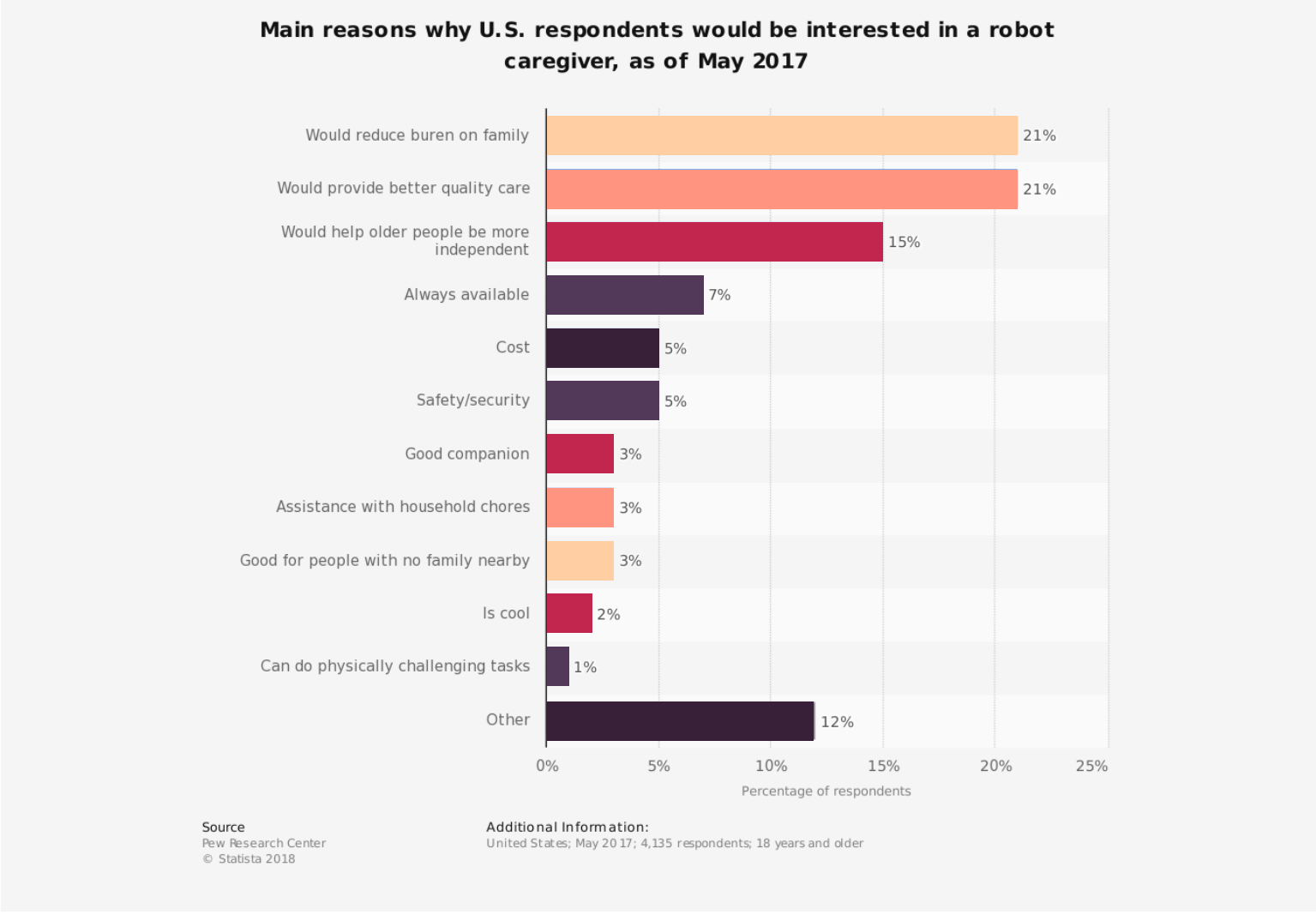
Top Manufacturers in the Global Healthcare Assistive Robot Market

- Cyberdyne
- Ekso Bionics Holdings
- GaitTronics
- Hansen Medical
- Hocoma
- HONDA Motor
- Interactive Motion Technologies
- Kinova Robotics
- KUKA Robot Group
- ReWalk Robotics

(Mendez, 2018)

Manufacturers of companion robots are lobbying hospitals and doctors to switch by detailing the benefits of companion robots in the healthcare space. These benefits include less workload on nurses or caregivers, happier and more empowered patients, and more independent patients. One article cites three reasons the elderly can benefit from service robots in their daily care. These three reasons they detail throughout the paper are assistance, monitoring, and companionship (Sharkey, 2012). Another source enforces the idea that these service robots are not supposed to replace a human connection, but facilitate the patient's independence. This loss of independence and loneliness due to isolation is a problem for older people. Therefore, in cases of a caregiver shortage, this switch to a robotic caregiver can help the older patient as well as the facility the caregiver shortage originates from. Using the idea of service robots as caregiving supplements instead of replacements is a main idea that company's such as Joy for All's companion pets and the robot ElliQ from the service robotics industry employ (Wiederhold, 2018).

Empowering patients through these innovations in the medical device industry is also a major benefit for doctors and hospitals to switch to robotic assistive technology. Elderly patients can become more independent and happier about their conditions through the use of robotics such as the exoskeleton suit for walking assist purposes. Increasing psychological welfare with these technologies allows for faster recovery as well as happier patients, which can ultimately help doctors hoping to discharge their patients in a timely manner (Sharkey, 2012).



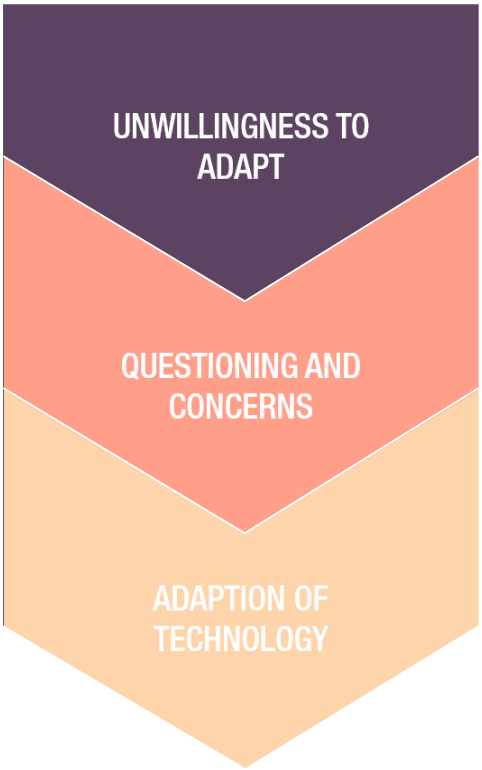
(Pew Research Center. n.d.)



(Campbell, 2016)

Concerns

Technology is an ever-advancing construct used and developed globally every single day. The landscape these mechanics present leave many concerns in each of their respective fields. In a similar circumstance one might question, if the smartphone they use multiple hours a day will have an adverse effect on the user in the future. Multiple concerns in this regard also arise in the medical field, specifically in the use of robotic companionship in providing medical care for the elderly. Comparable to the smartphone case mentioned previously, we must dichotomize each matter carefully to understand the specifics. In the next few parts we will discuss many of these apprehensions to this new technology and how best to interpret the challenges ahead.



LEVEL OF ROBOT AUTOMATION & REAL LIFE EXAMPLES

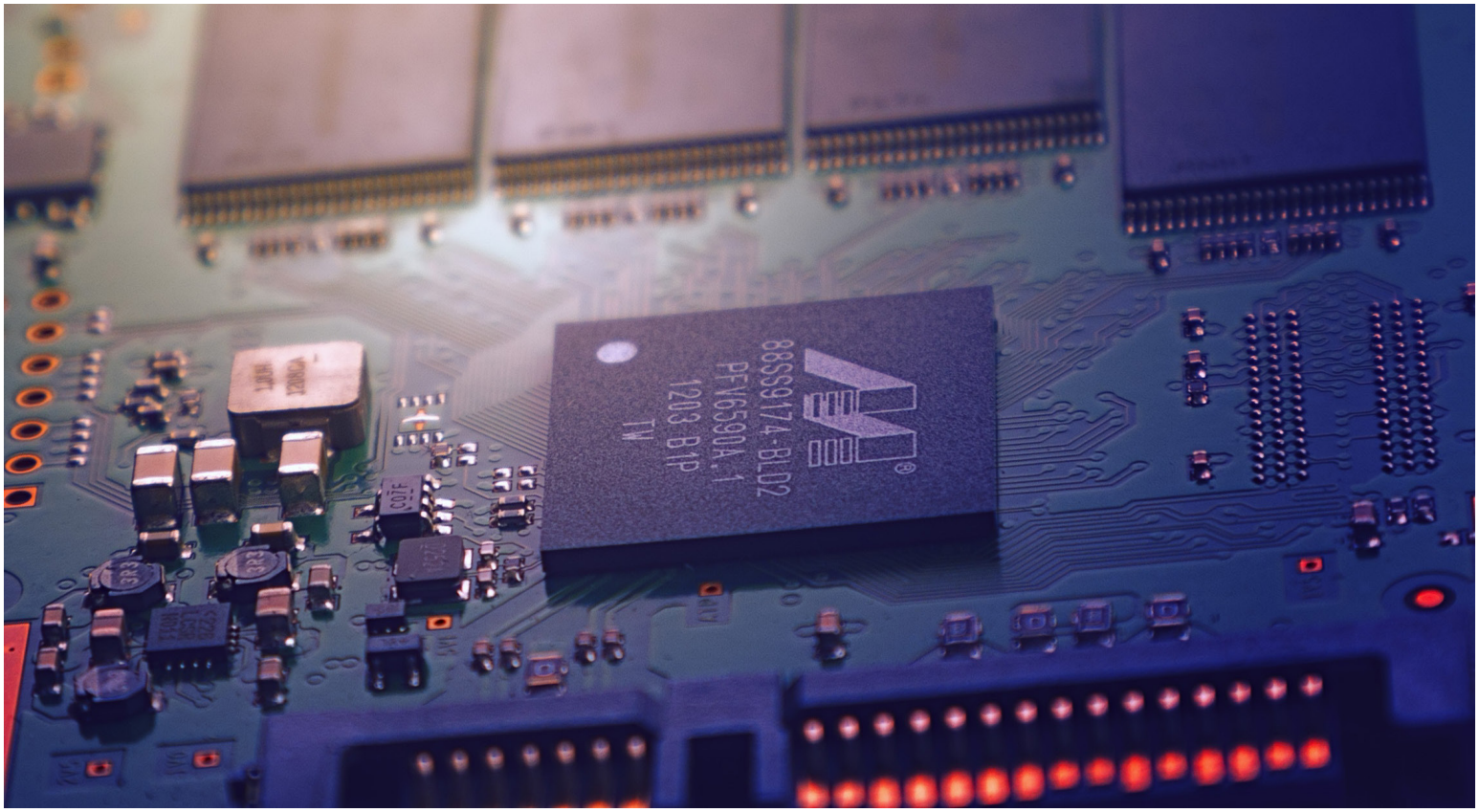
Currently, most robots are not yet completely autonomous and therefore need a human component to operate. Assistive technology robots in particular face many challenges because they must overcome an incredibly large number of variables. In addition, there are different levels of a robot's capabilities (see table below). Many robots can accomplish a specific task autonomously, such as handling objects or navigating a house, but the robot itself is not completely self-sustaining (Shishehgar, Kerr & Blake, 2018). However, certain robots, like telepresence robots, utilize a mixed approach of robot abilities supported by human capabilities in order to address a wholistic solution (Michaud et. al, 2007).

Fully autonomous machines are not far away, yet when the technology arrives, society must decide how to apply the law to machines and assign liability should they malfunction. For example, self-driving vehicles are already on the road, and fully autonomous or driverless vehicles may be here in five years or less (Bruyne & Werbouck, 2018). While the cars may have the capability to drive by themselves, society may ultimately decide not to legalize them because self-driving cars may have to make moral decisions: should the car swerve to avoid pedestrians and risk its occupants or protect its occupants and hit the pedestrians (Indurkhya, 2018). As discussed below, these moral questions only become more complicated when applied to the medical field. Until humans can fully trust autonomous robots, the machines may not realize their full potential.

Levels of interaction in assistive robotics

Level	Example	Physical Interactions	Spatial World	Safety
0	Medical monitor	Only communication	None	Level of static domestic appliances
1	Floor cleaning robot	Accidental collisions	2D	Level of power tools & vehicles
2	Object retrieval robot	Transfer of objects, holding tools or objects, providing reactive forces	3D	Additional user protection required
3	Wheelchair robot	Active movement of user's whole body or parts of body-holding tools or objects, providing reactive forces	3D	Safety critical & design essential

(Adapted from Meng & Lee, 2005)



(Plenio, 2017)

CHALLENGES WITH AUTOMATED ENABLED ASSISTANTS

Overall, the greatest challenge is how to develop effective technology and get it into the hands of the elderly. This challenge is amplified by the intense demand for elderly care in the current market. The overall challenge can be broken down into four key problems:

1.) **Assistive technology is underdeveloped for the tasks it's needed**

Up to date, the greatest progress has been focused in the industrial sector of robots, which focuses on a machine's accuracy and precision. In contrast, assistive technology requires high task flexibility to be able to function effectively in a domestic household (Meng & Lee, 2005). The assistive robot must also be able to navigate a home's changing conditions (stairs, doors, carpet, etc.), protect its owner's safety, and protect its own safety (Michaud et al., 2007).

2.) **Current technology doesn't necessarily address the real needs**

While assistive robots can accomplish impressive technological feats (i.e. cleaning, fetching objects, monitoring its patient, etc.), the robot's ability may not effectively address the patient's actual needs (Michaud et al, 2007).

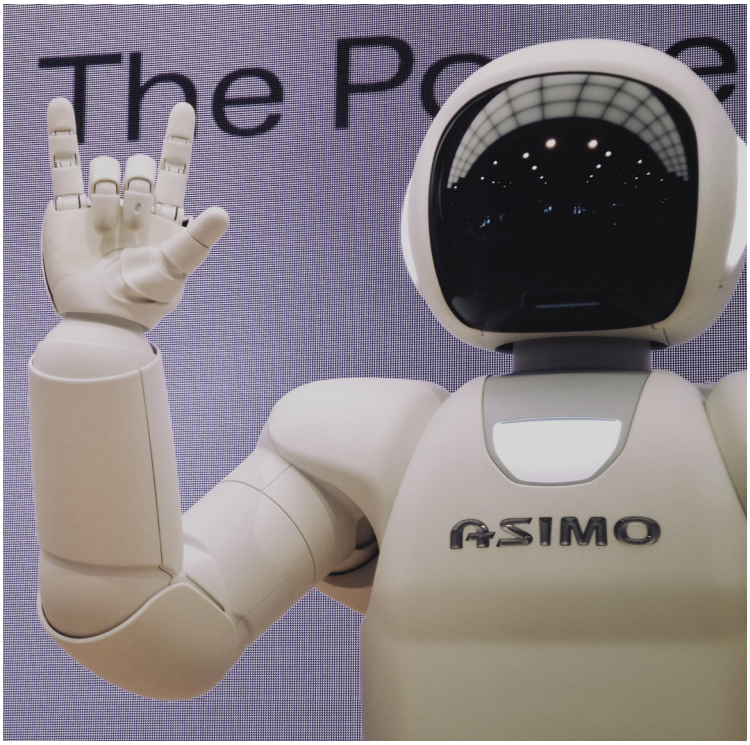
3.) **This technology is difficult to access due to high costs**

Regardless of the robot's ability, it will fulfill the pressing demand of elderly care if the potential consumers cannot afford it.

4.) **Elderly users may not be able to use/ understand the technology**

The target market of elderly will most likely not be familiar with advanced technology. Therefore, the tech must be as minimalistic and as intuitive as possible (Meng & Lee, 2005).

DOES MACHINE AGENCY SUPERSEDE HUMAN AGENCY?



(V., 2018)

With current technology, robots can make decisions in place of humans, but whether they should do so is an infinitely more complex topic. Humans already trust machines to perform repetitive tasks in the industrial world, but as robots become increasingly autonomous, such as making medical diagnoses or beating humans in chess, humans must decide if machines should be allowed to make moral decisions (Bigman & Gray, 2018). This decision is especially difficult because as a society, individuals differ on what is considered right and wrong. Therefore, robot manufacturers must decide what qualifies as a legitimate moral decision.

One side argues that while a machine can compute complex logical decisions, ultimately machines lack the agency (including traits such as interactivity, adaptability, moral reasoning, etc.) required to make moral decisions (Bigman & Gray, 2018). The machine itself lacks autonomy because it can only act in a way that it has been programmed (Banks, 2018). There is also always the potential for human error. In one experiment, Twitter released a chatbot to communicate with users and answer questions via "repeat after me" programming. However, Twitter users intentionally exploited the design and the chatbot turned vulgar, racist, and other undesirable traits (Banks, 2018).

The other side essentially suggests that machines could potentially make decisions equal to, if not better than, humans. They argue that human decision-making can be distilled into an algorithm, such as how to play chess. Yet, when that algorithm is put on a machine, humans no longer trust it, even when the results prove more reliable than human decisions (Indurkhya, 2018). In addition, those who are against moral machines cite that machines are susceptible to hacking, yet those in favor of moral robots argue that humans can also be essentially "hacked" via bribes and blackmail (Indurkhya, 2018).

Society accepts machine agency to a certain level of decision-making, but once a human life is at stake, the decision should remain in human hands.

However, neither side wins if the general population refuses to accept moral machines. Currently, research suggests that society accepts machine agency to a certain level of decision-making, but once a human life is at stake, the decision should remain in human hands (Bigman & Gray, 2018). Therefore, the use of moral machines in checkout nurses and companions in elder care may not be widely accepted.

RESISTANCE TO ADAPTING THE TECHNOLOGY

As with many technological advancements, there is a need to outline a clear defense of why those may hinder change. A SWOT analysis can be used to define these challenges, so that they may be applied. In finding the strengths, weaknesses, opportunities, and threats of this newer type of medical care we may adapt. Resistance to these medical robots can be signified as a threat to those in the medical field in the following ways:

1.) Should professionals in the medical field be at fear of losing their jobs?

2.) Are there security and privacy concerns we must understand?

The first question above can be answered by allowing those to understand, as current technology stands, these robot companions that provide hospitality amongst other things to seniors are mostly assistive devices. Meaning, the technology fosters better experiences for those they help by a mutual or symbiotic relationship that can only further the help those in the medical field wish to provide.

Lastly, security and privacy concerns may also have an affect on the struggle to adapt technical progress. In the age of privacy, we are led to to believe there must be those that are consumed by sharing every aspect of their lives or those that reside in the shadows. In reality it is mostly on a spectrum that most people find themselves on as a matter of convenience. Likewise, we must find a middle ground between information that is collected by these assistive devices, while also allowing it to develop and better equip itself for care. The rising heartbeat of an individual might predict this, or in the case of a stroke, this could be circumvented by machine learning. There is little need for worry because all these rising concerns may be cured.





Top: (Rawpixel, 2018), Bottom: (Dahan, 2017)

Moxi: The Robotic Nurse Aid



(Diligent Robotics, n.d.)

The idea behind Moxi was to find a way to aid hospital staff, but not replace them.

In September of 2018, the company Diligent Robotics, released their newest product, Moxi: a robot built to aid nursing staff. According to the Diligent Robotics Communication Team in an article for DigitalTrends.com the idea behind Moxi was to find a way to aid hospital staff but not replace them. Although currently the market is being bombarded with robots that can move patients or aid in surgical performance, Moxi is there for clinical work so that nurses can spend their valuable time with patients (Diligent Robotics, n.d.).

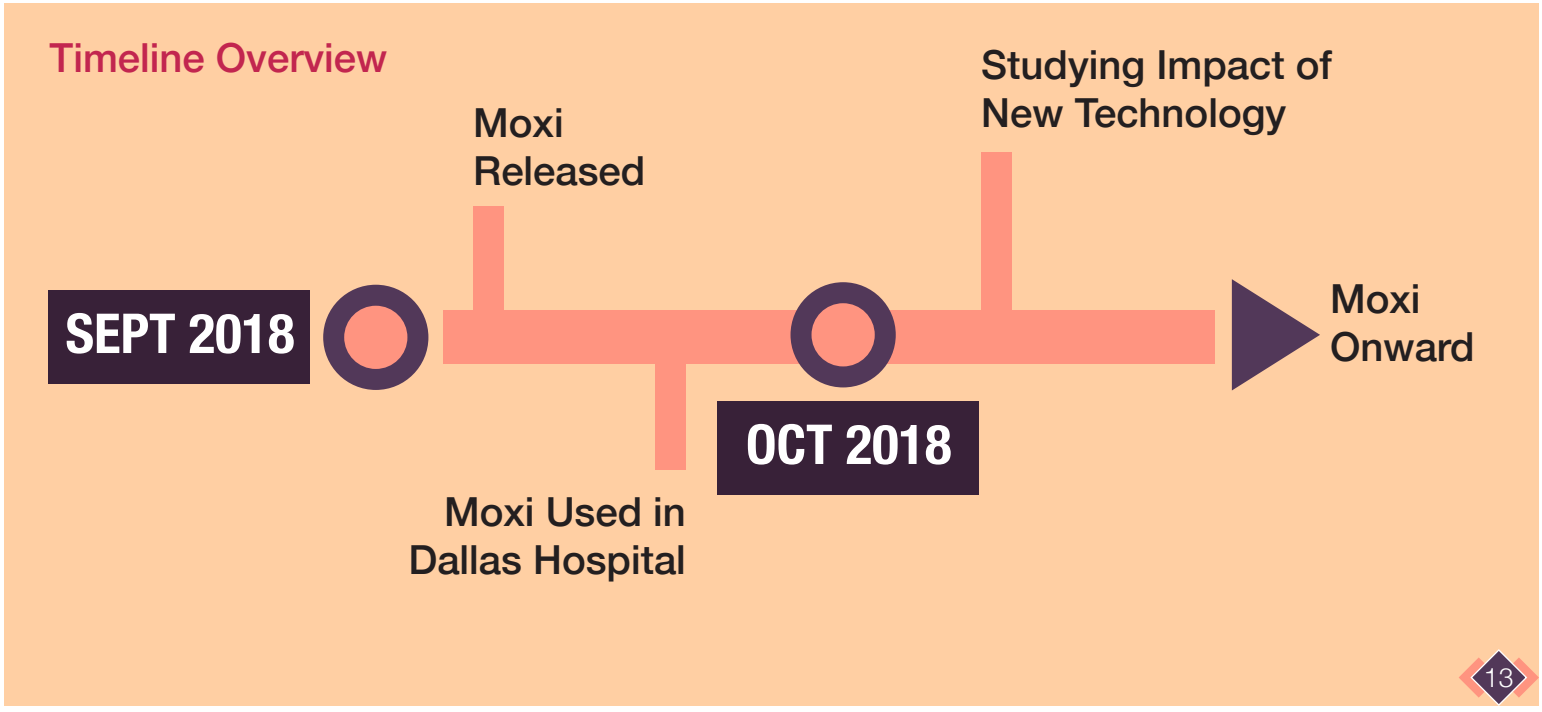
According to the Diligent Robotics team website, Moxi is a socially intelligent teammate, who works autonomously. It travels on its own by having a hospital map installed into its software. To combat safety issues while navigating often busy halls, the robot has sensors able to detect a human presence or path obstructions (Diligent Robotics, n.d.). Using the long, jointed arm, Moxi is able to reach objects in high places or place items where they can used by hospital staff. To make the robot feel more inviting to work with, the makers designed a face for it that would light up with different emotions, head movements, and a voice. It is able to nod its head in agreement when questioned and will have light up hearts show up on the face screen when it is experiencing something positive. The small things like this, as well as the greeting Moxi is able to give when passing to its human companions during work, have created the feeling that Moxi is more than just another piece of technology for nurses, it is a co-worker (Diligent Robotics, n.d.).

INTRODUCTION TO HOSPITALS & FUTURE OUTLOOK

The first hospital where this robot was introduced was the Texas Health Presbyterian Hospital in Dallas, in the Neurology and Stroke Unit. This was the first time in the U.S. that a hospital had utilized any kind of robot technology, run by artificial intelligence. The creators of Moxi, as well as the hospital employees hope that by implementing this technology early on, they will not only be more efficient, but also be able to find ways to improve it based on their experiences (Mogg, T., & Mogg, T., 2018, September 19). The testing will be done in two separate phases. The introductory phase started in September 2018 focusing on integration and demonstration of the robots use. At the beginning of October 2018, the hospital and Diligent Robotics team turned their focus toward studying the new technology's impact. A registered nurse (RN) at the hospital, Phobe Lype, stated in an informational video for Moxi's website that "Almost 30% of their [the nurse's] tasks are going to fetch things" (Diligent Robotics, n.d.). While patients will always need things such as a blanket or a new pillow to be brought to them, the nurses are able to delegate. They are then able to focus more attention on direct patient care. Since being tested, the technology has been slowly introduced into other hospitals throughout the state of Texas.

At Jennie Sealy in UTMB Galveston, the robot was introduced as one of the trial locations on January 21st of 2018. During its time there, it has been a huge help in processing logistics, handling supply deliveries and taking care of the small daily task that distract nurses from their primary tasks with patients. Nurse manager at the hospital, Ashleigh Thomas, notes that the robot has had a very positive response during its time there (Hennes, R., 2019, January 31).

Going forward, Diligent Robotics hopes to expand Moxi's presence into more hospital settings. Through the testing that has been going on over the months since it's unveiling they have been able to make small improvements that are only furthering the robot's abilities. In the future, the team hopes that continuous improvement of their robotic technology will further their ability to aid nurses and caregivers in the hospital setting. (Diligent Robotics, n.d.)



Part Two

GATHERING FROM DIFFERENT SHAREHOLDERS
ACROSS THE ASSISTIVE ROBOTICS LANDSCAPE



Robot Manufacturers

Key Insights

Based on interviews with representatives from Intuition Robotics and Kinova

- These technologies aim to decrease caretaker dependency, but increase social engagement with family and friends.
- Contrary to stealing people's jobs, robots may provide the solution to elderly caregiver shortages.
- Robotics manufacturers must ensure their customers' information remain private and secure.
- Cost of technology remains a considerable barrier to consumers.

OVERVIEW OF COMPANIES INTERVIEWED

The companies contacted and interviewed were Intuition Robotics, who manufactures the AI-enabled ElliQ, and Kinova Robotics, who manufactures assistive robotic technologies, without AI, but plans to move into the AI assistive robotics space in the future. The Intuition Robotics representative is named Danielle Ishak and her official title is User Experience Researcher. The Kinova Representative is Simon Latour and his official title is Product Manager of Assistive Technologies. Both manufacturer representatives expressed that they believe assistive robotics will help improve their customer's quality of life. Whether it is dealing with physically assistive technology, such as the technology offered at Kinova, or mentally assistive, such as the ElliQ offered by Intuition Robotics, robot manufacturers in the post-operative, assistive robotics field are dedicated to the ethics, security, and feasibility, both monetary and functional, of their products to make a positive difference in their users' lives.

ElliQ, from Intuition robotics, is manufactured to help aid with loneliness in elderly grandparents by enabling them to contact their loved ones easily via video chat. ElliQ will also set reminders for any aging loved one with memory loss. These simple tasks help family members of the elderly feel more secure that they're leaving their loved one better taken care of than before. Similarly, assistive technologies from Kinova such as wheelchairs and robotic arms help people, young and old, regain a sense of independence in their daily lives. This allows them to join their friends in normal activities and feel less lonely again. Although ElliQ utilizes AI technology while Kinova products do not, both assistive technologies present similar challenges and triumphs in the assistive, post-operative robotic technology space.



Figure 1.1 *ElliQ is a stationary robot that aims to increase social engagement and decrease isolation and depression.*

ETHICAL CONCERNS OF ROBOTS IN ELDERLY CARE

Introducing robots into elderly care raises several ethical questions: Will robots reduce a patient's level of human interaction? Will robots replace human jobs? From Ishak's perspective, these issues are unfounded. In regards to social interaction, she states "that's the purpose of ElliQ as a robot... We put a lot of thought in her design and interaction models to help increase engagement with family members." ElliQ utilizes today's numerous communication platforms, from video chatting to Facebook poking, in order to increase an older person's social life, while decreasing his or her feelings of isolation. However, the question remains: is virtual communication effective enough to compensate for the lack of face-to-face interaction?

"Nurses are highly skilled practitioners, where they can do almost everything up to prescribing medication or diagnosing a patient, but a lot of what they do in their job is ... dull, dangerous and dirty tasks." - *Danielle Ishak Intutition Robotics*



Figure 1.2 *Pitctured: Kinova's JACO RE Robot Manipulator.*

Robots that do not specifically focus on increasing human socializing, may unintentionally decrease human contact. At Kinova, Latour concedes that technology that helps the elderly become more independent, may decrease social interaction. However, he argues that in situations where, "there is already no social interaction... speaking with an AI, it's not like speaking to a human, but it's still better than speaking to nothing. If you're able to fill that gap, it's okay." In addition, if technology is successfully able to help older people be more independent, it may encourage them to spend more time outside the house with friends. "Instead of being taken care of, it's more like participating in the social life... that human factor needs to be there," Latour concludes. In regards to the job market, both representatives whole-heartedly agree: people should not be worried about robots replacing humans or taking their jobs. On the contrary, the elderly health care sector is somewhat unpopular and it already faces a severe shortage of help. This is congruent with our previous research which cites a healthcare crisis in which the aging population needs more help than the current healthcare system can offer. (Sharkey, 2012)

Ishak explains that robots have great potential to aid health practitioners. "If you look at nurses in a hospital for example, nurses are highly skilled practitioners, where they can do almost everything up to prescribing medication or diagnosing a patient, but a lot of what they do in their job is... dull, dangerous, and dirty tasks." In reality, robots could be logging health information or cleaning patients nurses can focus on highly skilled tasks that only a human can do. While Latour believed robots may be capable of replacing humans one day, that day is a very long time away. Right now, the industry should focus on making robots that can perform tasks that create a better work environment for caregivers, as opposed to replacing humans altogether. Furthermore, Latour explained that Kinova currently does not manufacture fully autonomous technologies due to the complex moral considerations.

SECURITY, PRIVACY, COST & FUTURE CONSIDERATIONS

Both representatives are fairly confident that their company can provide adequate security. Kinova is especially safe because the assistive technology they manufacture does not store customer information like the ElliQ at Intuition Robotics does. Therefore, Ishak discussed security in further detail. She explained that Intuition Robotics used a combination of cybersecurity tactics and de-identified aggregate data, so that they can run analytics to improve ElliQ, while still protecting their customer's privacy.

Concerning privacy, Ishak also mentioned that the older population is especially sensitive to being monitored. However, adult children often purchase ElliQ because they know their elderly parent needs supervision and could easily be swayed to give out information. The adult children may feel guilty by leaving an elderly parent alone or living far away. So, ElliQ attempts to strike a balance by providing "a very light way to give the child enough information that they feel at peace and comfortable with their adult parents aging remotely and it's not super invasive."



Figure 1.3 *Older adults are especially wary of being monitored. (Hermant, 2018)*

The cost feasibility presents a considerable barrier for both manufacturers. Intuition Robotics is currently working on subsidizing ElliQ through insurance companies, but they anticipate that the overall cost will decrease through economies of scale. At Kinova, Latour also agrees that the cost of technology will decrease as more people buy the product. In addition, they are conducting studies to support that their products are more cost-effective over the long-term. However, until more people adopt either technology, costs will remain high, especially because of extensive research and development costs.

Cost Estimate:	KINOVA *JACO RE Arm Manipulator:* $50,000	ELLI·Q $1,499.99

Another difference between the two manufacturers was their ability to be implemented into hospitals. Intuition Robotics believes there is a future where ElliQ is adopted into hospitals and it will be feasibly incorporated into rehabilitation practices for the patient recovering from a surgery. While Kinova products may not be utilized in hospitals, they are feasible for rehabilitation as well, and helps to restore a quality of life in helping people move their arms again. This can even restore their ability to do a simple task such as feed themselves.

Both robotic manufacturer representatives perceive that current user opinion is positive. In the case of Kinova, their products were originally created by their CEO who saw a need from people in his daily life for these assistive robotic technologies, so their products are entirely inspired and created based on user opinion. This leads to high user involvement in the research and development of their products, and high user satisfaction. In the case of Intuition robotics, their user opinion is mainly measured from beta market testing and this contributed to the startup's credibility moving further with the full launch of their product.

Doctors and Care Providers

Key Insights

Based on interviews with doctors and care providers of varying specialities

- When compared to the rest of the population, there seems to be more resistance amongst elderly care patients when it comes to technology.
- In today's day and age, doctors and care providers must know how to use the latest technology in order to stay ahead of the curve.
- Both robotic technology and new technology may cost more up front, it is actually cheaper for the patient in the long run.
- Even with the rapid progression of technology in the medical field, HIPAA has been able to give care providers the guidelines necessary to protecting patient privacy.

CARE PROVIDERS: TECHNOLOGY DRIVES ADVANCEMENT

The doctors and care providers surveyed were from a wide range of areas. From elderly care providers, to orthopedic surgeons, to dentists, the data compiled was able to give us a wide range of opinions about the integration of technology in medical care in general as well as in their specific fields of study. All of the interviewees were asked about their personal interactions with modern technology, their patients interaction with whatever technology they may be using, and any potential concerns these doctors may have about the progression of technology in the future. While the specific answers varied, there was a common theme among the majority of the interviews. This theme is that technology has allowed their respective field to progress drastically in the past 20 years while improving the overall experience for the patient that they are trying to help.

The first part of the questionnaire focused on their interactions with technology, whether robotic or not. If they don't have robotics in their facility, we asked how they would like to see them integrated, if at all. Additionally, they were asked about technology maintenance policies and if they think technology has been a positive or negative in their experience with healthcare. Though all of our doctors and care providers interact with technology, the level of sophistication varies from person to person. There was also somewhat of a divide between the frequency of interaction with new technology. As an orthopedic surgeon, Dr. Peter Sabonghy works with robotic equipment every day. When asked how much time he is willing to devote to learning this new technology that is so prominent in the field of surgery, he says, "I spend 100 hours a year training for new technology whether it's training or labs but it's important to me. I'd guess the minimum is 10 hours a year. It varies from doctor to doctor but I think it's important to become confident in the technology you're using". This was a common sentiment among our interviewees, whether they took initiative or thought of it as a necessity. When asked about his devotion to learning new technology, Josh Richard, an urgent care nurse practitioner said, "Oh as much as I need because it helps me provide for my family." So while some may arrive at the same conclusion, they take different ways of thinking to get there. However, it is apparent that to be successful in the medical field today, it can be necessary to go above and beyond what is required to stay ahead of the curve.

PATIENT'S INTERACTION WITH TECHNOLOGY

The second section of the questionnaire focused more on the patient's interaction with the technology used by these care providers. Questions ranged from patient trust, to knowledge of the machines, to satisfaction. The overwhelming majority of respondents said that they feel as if patients trust the technology that is being used on them. As technology has become more and more common in everyday life, consumers trust and familiarity with being around a lot of this technology that has made its way into medicine has increased. However, while trust has increased, that has not necessarily translated to patients knowing what these newer machines do.

"They don't want to learn technology and thought that we were spying on them because some of our technology has cameras." *- Brandi Johnson, Home Instead*

While the younger portion of the population is mostly eager to try and learn about this new technology, the older portion feels somewhat of a disconnect. Brandi Johnson of Home Instead, a senior care company that sends caregivers into residences as opposed to assisted living, had some interesting thoughts on why seniors might not trust or want to learn about both robotic and new technology. When asked why, she responded, "They don't understand it. They were nervous that something was going to go wrong, which is pretty much impossible. There's just a fear. They don't want to learn technology and thought that we were spying on them because some of our technology has cameras"

Quotes like these seem to indicate that much of the resistance to new technology doesn't come from the majority of the public, but a portion of the population that is older and maybe less willing to adapt to the world around them.

Rosa Mata, a nurse administrator at Brazos Valley Urgent Care summed up this thought nicely by saying, "I think it's more of an age factor. Someone in their 40's would probably understand quite a bit of it. Someone in an older generation like their 70's or 80's would be lost". This is not to say that everyone in older generations is incapable of both learning about new technology and how to trust it. Providers especially in elderly care and surgeries have a challenge on their hands in educating the older public on their machines.



Figure 2.1 *GrandPad is a tablet created for seniors*

When Brandi Johnson introduced the "Grandpad," and iPad for seniors, at her Home Instead locations, it did not go over well. "They felt it was an invasion... they thought we were going to be spying on them," she said. Johnson felt a big part of their apprehension was because the seniors felt they would do something wrong, "They're just scared. they didn't know how to do it. They didn't want to learn the technology." However, Johnson continued to say they did have more success with residents who were younger.

CONCERNS WITH TECHNOLOGY MOVING FORWARD

The last section of the interview conducted was over the care providers potential concerns with the ever-evolving technology that has seemingly taken hold of all sectors of medical care in the past 20 years. These concerns related to higher costs or insurance premiums for the patients, any disadvantages that newer technology may present, and patient privacy concerns. While many of the nurses interviewed did not have the necessary information to make a solid statement on the cost for the patient, a few of the respondents had an interesting take on the effect of high-tech machines and cost to the patient. Both Sabonghy and CJ Wilson, a nurse practitioner here in College Station echoed similar thoughts, saying that while new technology can obviously cost more up front for the patient, in the long run it can actually come out to being cheaper. With advances in technology it has become easier for doctors to diagnose and treat patients problems with a decreased risk of error. This prevents the need for more procedures in the future and a greater chance at a healthy patient after one appointment.

The answers for disadvantages varied, but one of the more important responses had to do with doctors that may come in contact with robotic technology. While these machines have progressed so much and help doctors a great deal, they may cause some doctors to become over reliant on the technology, and try to fill the void between their skill and what the patient needs. Many times, this is not what the machines were designed to do and if they cannot sufficiently fill that void, then it leaves the patient in a bad spot.

Patient privacy is another potentially big concern as machines and offices become more interconnected than ever. All respondents said that their facilities take measures to be certain that their patients information is stored in a secure location. The big rule system that these providers follow is HIPAA, which governs how they are allowed to store and protect information. This gives providers a standard way of storing and keeping information out of the hands of unwanted third parties.

Ensuring Data Protection:

With her experience as the Clinical Director and Director of Nursing for a surgery center and outpatient surgery center, Debbie Northway explained the lengths centers have to go to in order to protect patient data. “We had cyber security companies that would come in and assess our risk and make sure that we had the systems in place to protect it,” she continued. Every piece of data is valuable. Northway asks, “When patients check in, and you’re waiting room and writing their name on a list, how do you protect that?” What does that mean for robotic technologies that could collect even more personal information?



Finally, respondents were asked if their facility had a contract with a certain manufacturer. While many said that they were unaware if a contract existed, there was one answer from Dr. Gordon that stood out. He said that while his facility does not have a contract with one manufacturer, it is important to him to have the majority of his equipment from the same company. This allows his practice to have the peace of mind that should anything short-circuit or stop working, they know exactly who to call and how fast they can address the issue.

Patients

Key Insights

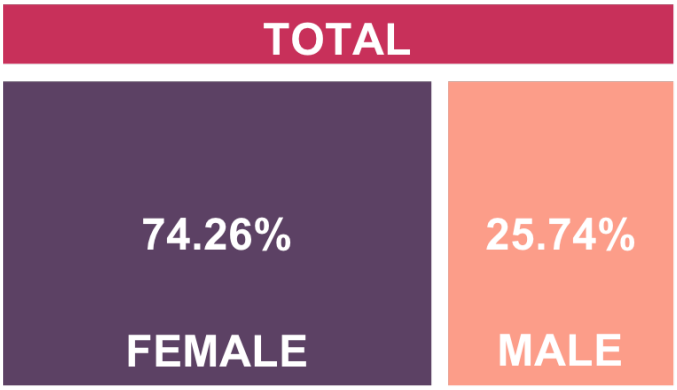
Based on survey results from 125 patient respondents

- A majority of patients in a hospital setting believe that there could be improvements to their experience with time and service.
- Most people surveyed have not had experience with robotic technology in a medical setting, but have had at least a little experience with smart technology in a medical setting or their daily life.
- Although new technology can be uncomfortable for some, most people are willing to try it if it has potential to make their life easier and is reliable.
- Respondents preferred a robot with more human features.
- Despite fears from confidentiality risks, malfunction, and less human interaction, 66% of those surveyed believe that having robotic technology in a nursing home setting could be beneficial.

SURVEY DEMOGRAPHICS & MAJOR CONCERNS

Our survey, created on Qualtrics, was taken by a total of 125 respondents over a one month period. Out of these responses only 122 were valid based on level of completion. The survey was distributed to people who had recent experience as a hospital patient, nursing home residents, and family members of nursing home residents who were not able to take the survey themselves. The survey consisted of 16 questions and on average took respondents 3-4 minutes to complete. Our team was able to gain responses through sending the survey on Facebook, through email/text message, and in person.

Holding 29.70% of responses, the majority of respondents were under the age of 50. 27.72% of the responses came from persons between the ages 50 and 59, 18.81% from between the ages 60-69, 15.84% between the ages 70-79, and 7.92% were above 80 years old. 2.46% of respondents lived in a nursing home and 97.54% of respondents lived outside of a nursing home. 74.26% of those who took the survey were female and 25.74% are male.

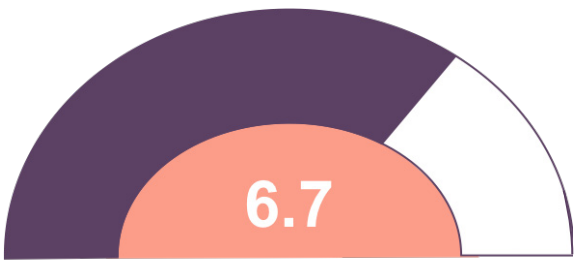


The survey starts off asking survey takers what their major concerns are for those around them as they age. They were given the options of companionship, keeping up with medication, nutrition, staying active, memory and other (open response). They were able to choose all that they felt majorly concerned them. Out of these options and the write in responses, the biggest worries for our group surveyed was staying active with 71 out of 122 responders choosing it. The second most chosen concern was memory with 67 out of 122 respondents choosing it. Companionship was determined to be the 3rd biggest concern, keeping up with medication being the 4th and Nutrition being the least worried about issue. 16 out of the 122 respondents wrote in responses in under the “other” category; most of which had to do with finances.

CURRENT INTERACTION WITH TECHNOLOGY

The second question was “Under the current medical system, do you feel that your medical needs are met in a timely manner?”. The responses were set on a scale of 1 to 9, with 1 meaning not timely at all and 9 being extremely timely. The mean response out of the 89 answers was 6.38 with a standard deviation of 2.16 and the mode was 7. This means that although most respondents felt that their services were done in a timely manner, there is still some room for improvement.

Next the survey asked respondents the level of comfort they have when interacting with new technology. The scale ranged from extremely comfortable to extremely uncomfortable and gained 99 responses. 68.68% of those responses answered as either moderately comfortable or extremely comfortable. 19.19% of responses were neutral of only slightly comfortable and only 12.12% of responses cited any discomfort operating new technology. The following question asked about the level of trust they have that technology will function as designed. The answer options were laid out on a scale from 1 to 9 with 1 being a strong lack of trust and 9 being a strong trust. Out of the 81 responses the mean was 6.7 with a standard deviation of 1.64 and the mode was 7.



In the following questions it was found that 76.24% of those surveyed had never had experience with robotic technology,which is defined as the use of machines to perform tasks done traditionally by human beings, in a hospital/medical setting. It was also found that 66.34% of respondents interacted with smart technology everyday and 10.89% of respondents rarely or never interacted with it. Smart technology was defined to them as an electronic device, generally connected to other devices or networks that can operate to some extent interactively and autonomously. The major smart technology interactions had to do with Alexa primarily and then Google home, smart phones/watches and Tile.

“It just depends.....I’m skeptical and have the expectation of serious testing and documentation of results prior to release.” *- Respondent in regards to the reliability of robotic technology*

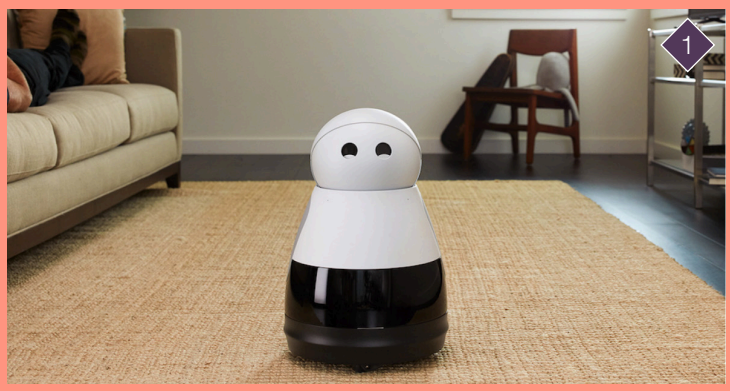
With the survey we were also able to gather that a majority of users felt that robotic/smart technology is reliable. The major concerns of users had to do with malfunctions and having less human interaction. The other less prominent answers chosen (in order of highest chosen to lowest) were loss of human jobs and higher insurance premiums. When given the option to write in responses most wrote in having issues with confidentiality. When asked if they felt that robotic technology was a risk to their physical/ informational safety most were unsure or leaning toward it being a danger.

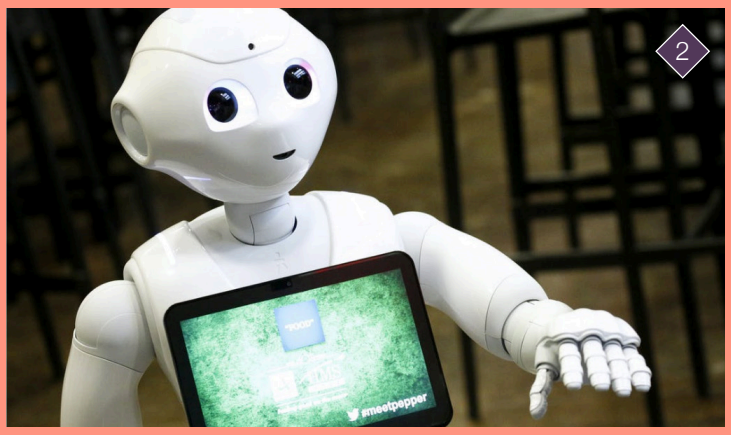
The survey proceeded to ask if the responders felt that robots are a luxury in the medical field, the responses were mixed. Although we got the highest response for the answer of probably yes with 30%, the response rates for definitely yes, might or might not and probably not were following close behind all having responses percentages between 18% and 26%.

66%

... say yes to introducing robotic technology into nursing homes

When asked out of the following (pictured) which was most appealing choice 2 was chosen. This was asked to get an idea of what patients may be more comfortable with when facing future possible interactions.







Finally it was asked if the survey audience felt that it would be helpful for robotic technology to be introduced into a nursing home setting. 66% of responses were yes, robotic technology would be a positive introduction in possibly helping with medication reminders, deliveries and clerical aid. Those who were unsure cited elderly discomfort with the new type of interaction and felt that it may be better with clerical activities leaving nurses to primarily deal with patient care.

"I think there will be members of my generation that will be afraid, future generations though will see robotics as normal."

- Respondent on robotics in nursing homes

Part Three

CONCLUSIONS AND RECOMMENDATIONS: THE FUTURE OF ASSISTIVE ROBOTICS



Future of Assistive Robotics



Our study consisted of qualitative interviews with doctors and robot manufacturers, and a survey distributed to a sample of patients. Through this, our study was able to gain 3 different perspectives regarding the future of companion robot technology and experiences regarding current technology use.

PERSPECTIVE ONE: ROBOT MANUFACTURERS

We started off by contacting robot manufacturers: Intuition Robotics and Kinova Robotics. When interviewed, these robot manufacturers had an extremely positive outlook on the new technology being placed in environments such as hospitals. They look at technology as something that will be a huge help for patients and nurses, especially in regard to companionship and aide. Though they see the future of the medical robotics as being a success, the manufactures still have to place a lot of their focus towards risk mitigation and facing ethical dilemmas. Going forward they specifically feel that a huge worry for customers is the fear of information leakage and taking away some social aspects of people's lives when they are interacting with technology instead of people. Knowing this, they have tailored their designs to going forward to include screens that can allow patients to be in contact with real humans when needed, and installing improved systems for data protection. They believe their robots will especially be able to aide in nursing homes, where patients often need a lot of day-to-day aide. In order to achieve robotic expansion into nursing homes, their technology will need to be user friendly for the older population that isn't as adept to technology and address consumer needs such as rehabilitation.

PERSPECTIVE TWO: DOCTORS & CARE PROVIDERS

Next, we interviewed doctors and care providers to get their insight on how caregiver technology has influenced their practice. Across the board, they see introducing new technology as a necessity when it comes to keeping up with medical advances. Most doctors go through extensive, yearly training to keep their medical practice up to date, so learning new medical technology trends is a part of their job expectations. Their biggest resistance in introducing this new technology has been with the elderly population. It could be very challenging to bring robotics into a nursing home setting, due to the unease of its residents. To avoid this mistrust in the future, slow introduction will be extremely helpful. They could start off with smart technology that sends reminders, and work up to having robotic guidance around the facility. While having elderly patients directly interact with robots is a big challenge, nursing home facilities could bring in robotic aids, such as Moxi, to help with the more mundane, behind the scenes tasks that nurses must do. This would allow for more face to face interaction between the patient and care provider.

PERSPECTIVE THREE: PATIENTS

Finally, we distributed a survey amongst a sample of patients to reflect on their experience with technology in medical settings and their feelings towards it. Overall, most patients feel that their experience in medical settings could use improvement. After running this study, our team feels that, not only would introducing robotics help in efficiency, it would give a more individualized patient experience in which doctors and nurses could spend more time focusing on the patient. One of our respondents wrote in that “I think there will be members of my generation that will be afraid. Future generations though will see robotics as normal”. This quote reflects many of the responses our study received. Most patients see robotic care as having a place in the future of medical care, but worry about understanding it when it isn’t a part of their life already. Reflecting on many respondents seeing resistance from the elderly population, it can be assumed that the introduction of this technology will be a slow process. Many facilities will most likely face resistance when robotics are first introduced but over time, once things settle and it is a normalcy, the robots could be a huge help to the staff and patients.



Looking Forward: Robotics Long-Term

Based on the findings from these three perspectives, it can be predicted that robotics hold a place in nursing homes of the future, but it will need to be slowly implemented and tailored to elderly care needs. From all points of view, there is an understanding that people are nervous about new technology, but in the long run it is worth it to have a higher standard of living.

Next Steps: Recommendations

FOCUS: PROVIDE TECHNOLOGY TO THOSE WHO USE IT

While current robotic technology is incredibly innovative and it holds great potential, it faces the most resistance from the very market it is trying to help: seniors. In theory, these technologies will improve lives, but in practice, it becomes more complicated. The majority of elderly people find it difficult to learn new technologies, they are afraid that robots will malfunction, and they are suspicious that technology is spying on them and invading their personal privacy. Several care providers explained that even if older people are willing to learn new technology, it takes extensive time and training for them to understand it, even if that technology was specifically simplified for seniors. Even then, the seniors may not use the new technology consistently if they feel they are being monitored or they are afraid it won't work properly.

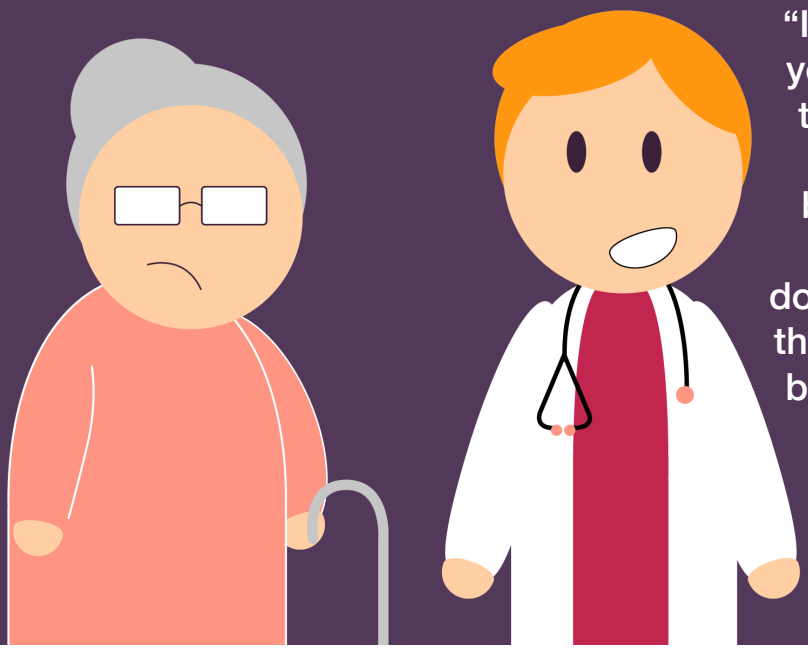
However, the doctors and care providers who take care of the older population are much more enthusiastic about the possibilities of new technology.

Doctors and caretakers generally agreed that they are willing to dedicate as much time as is needed to learn new technologies so that they can better serve their patients. Instead of being apprehensive, this segment is very excited about how new technologies can make their jobs easier and revolutionize the medical field.

By nature, people tend to become more set in their ways as they grow older. So, instead of trying to change the minds of the entire elderly population, we recommend that manufacturers focus on products that aid doctors and care providers in their daily tasks so that these professionals can focus more on taking care of patients. Instead of creating a robot that acts as a personal companion, manufacturers can focus on robots who help enter clerical data, make deliveries behind the scenes, and support nurses in other tasks that are, according to our interivew with Danielle Ishak from ElliQ, "dull, dangerous, or dirty." This strategy will help with the elderly care crisis in the shorter term.

"They don't want to learn technology and thought that we were spying on them because some of our technology has cameras."

- Brandi Johnson, Home Instead



"I spend 100 hours a year training for new technology whether it's training or labs but it's important to me... It varies from doctor to doctor but I think it's important to become confident in the technology you're using."

- Dr. Peter Sabonghy.

UTILIZING ARTIFICIAL INTELLIGENCE

In the case of Kinova, their assistive technologies do not currently utilize AI. Making changes to utilize AI in future technological innovations and in future R&D projects will help Kinova stay ahead of competitors and better serve their customers.

Implementing these technologies could present challenges such as those presented by those fearful of technology. Targeting younger family members who have a better understanding of how the technology works is a strategy that Intuition Robotics currently employs and that Kinova could consider going forward. Intuition Robotics finds that, in a lot of cases, the adult child of an elderly parent is the one who is making the purchase to help their parent. This purchase alleviates guilt in the child and helps the parent in their everyday life. This same strategy could be applied to Kinova robotics as they move forward with AI-enhanced innovations to their products and some elderly customers potentially become fearful of these technologies. Any company operating in the postoperative, assistive care robotics space should prioritize making seniors feel comfortable with the technology. However, technological innovations, such as implementing AI-enabled products, is essential in order to remain relevant in this industry's future.

ETHICS OF INTENDED USE & SOCIAL CONSEQUENCES

Some healthcare providers cite an over-reliance on the technology introduced to make their jobs and interacting with patients easier. Often, robotic technology is used to fill the void between the healthcare professional's skill and what the patient needs, which is not necessarily what the technology was designed to do. This could leave the patient feeling underserved and does not fulfill the goal of these technologies. Specific intentions for use need to be made more obvious by beta testing the technology in professional settings before introduction into the market to allow time to catch these possible gaps between professional use and patient needs.

Ethical concerns often arise because of the introduction of autonomous technologies into the medical professional space. Improved robotic technologies are often accompanied by a decrease in face-to-face human interaction. Manufacturers need to keep in mind the harm this decrease in social interaction can cause when testing their technologies. Efforts are needed to make sure the technology does not overstep any ethical limit of reducing human interaction for patients. Keeping social interaction as a high priority when testing and manufacturing assistive technologies is imperative because a decrease in human interaction is associated with depression and a decreased standard of living. Companies such as Kinova use their technologies to allow their users to increase levels of social interaction through their technologies by enabling them to complete normal tasks, such as eating and moving around, again. Intuition Robotics' ElliQ also enables this increased level of social interaction by allowing its users to more easily contact loved ones. Keeping in mind the ethical concerns of social interaction when introducing autonomous technologies is essential to the future of the assistive robotics industry because the user's health must come first.

Over the next 10-20 years, manufacturers can focus on developing products that are patient-facing. Consumers who are currently in their 40s to 60s are already more accepting of robotics and they are generally more technologically savvy. If manufacturers can continue to improve their current technologies to make them more cost-effective and reduce errors, then their target market will likely be much more willing to accept more personal robotic solutions.

Current State of Existing Technology

ROBOTICS HAVE YET TO ENTER THE MAINSTREAM MARKET

To gather our data, we conducted qualitative interviews with doctors and care providers, as well as a comprehensive survey to gauge patients opinions on the integration of technology into both elderly care and healthcare as a whole. Our findings across both the doctor and patient answers were relatively uniform. The majority of our patient respondents have not actually interacted with robotics in their experience with healthcare, while the doctors and care providers who were interviewed routinely interact with technology, there has yet to be widespread integration of robotics into facilities that do not conduct surgeries.

Before conducting these qualitative interviews and surveys, our group had the assumption that, like us, our respondents interact with some form of technology on a regular basis. While robotics are still making their way into mainstream use, both in daily life and the medical field, we knew that some practices have begun integrating robots to streamline the process for both them and their patients. This lead us to the expectation that doctors will have more insight into robotics over the patients that we surveyed. Over the past couple of decades there has been a drastic shift towards introducing new technology into the public. The introduction of these inventions have made life easier for everyone, but not without some reservations. There has been hand-wringing over security of information, spying by the government, and a general sense of caution as technology becomes smarter and smarter.

While we expected some resistance from the patients surveyed, there was less of an expectation of this thinking from the doctor and care provider side. When looking at the integration of technology into



different practices, it is easy to see why doctors would presumably have more enthusiasm with smart technology. Whether it's surgery, general check-ups, or elderly care, smart technology and robotics give doctors the opportunity to delegate some tasks to machines and see more patients while minimizing the chance of human mistakes. Obviously, the level of technology will vary from sector to sector of the medical field, but we felt that it was safe to assume that most of the doctors interviewed would react positively to the chance of better technology at their particular practice. Before conducting our interviews, our thought was that surgeons would be more enthusiastic and have a more intimate knowledge of robots while nurses and other care providers would support the idea of integrating robots, but would most likely have a relatively smaller understanding of how robots work and could be integrated into their facilities.

INTERACTION WITH TECHNOLOGY: PATIENTS

Like the almost all of the public today, the overwhelming majority of our patient respondents said that they interact with smart technology on a regular basis. The two devices that they had the most familiarity with were home companions that have been introduced in the past couple of years such as Alexa and Google Home. Since the majority of our respondents are not in assisted living (97%), it was plausible to assume that they would be familiar with this new technology that is making its way into homes today. Additionally, almost 70% of the patients surveyed were at least moderately comfortable with interacting with new types of technology. This makes sense when you consider that these new inventions, along with smartphones, are part of everyday life. What surprised us from the survey was that about half of the respondents considered robotics a luxury in the medical field. However, most agreed that potentially introducing robotic technology into nursing homes would be beneficial. Most patients were concerned with remembering to take medication, staying active, memory, and companionship as they get older. So when given the choice of having a robot to help them with many of these concerns, respondents were overwhelmingly positive.

70% of respondents indicated they were at least moderately comfortable with new types of technology

97% of respondents did not live in assisted living

ROBOTICS IN THE MARKETPLACE: SURGERY

After conducting qualitative interviews with various medical professionals, we found that the majority of those interviewed were pro-technology and would encourage the introduction of robots if given the chance. Multiple practices have already begun to use robotic technology in their everyday use, but mainly for surgeries and x-rays. This confirmed our initial thoughts, since smart robots are still in the process of integrating themselves into the marketplace. Additionally, the specific type of medical care directly impacted the care providers interaction with robotics. Doctors such as surgeons and dentists are essentially experts in the robotic field while nurses and elderly care providers have significantly less experience with robotics. Overall, while certain fields of medicine interact with robotics more than others, everyone in the medical field interact with technology on a daily basis, and have become reliant on it.



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